INDIAN STARTUP FUNDING ANALYSIS

2015-2020

SQL QUERIES

MySQL queries answering the 6 key questions from the problem statement

- Table: cleaned_startup_funding
- Columns: Date, Startup_Name, Industry, SubVertical, City, Investors_Name,Investment_Type, Amount_USD, Year, Month, YearMonth, Investment_Type_Clean

Total funding trend by year and month

```
SELECT
    Year,
    Month,
    SUM(Amount_USD) AS total_funding,
    COUNT(*) AS total_deals
FROM cleaned_startup_funding
GROUP BY Year, Month
ORDER BY Year, Month;
```

OUTPUT :-

Year	Month	total_funding	total_deals
2015	1	370171768	40
2015	2	394262600	42
2015	3	463327000	47
2015	4	981000000	52
2015	5	326185000	44
2015	6	536994000	58
2015	7	1606657000	70
2015	8	1063816000	65
2015	9	1470520000	61
2015	10	254961000	59
2015	11	899275000	62
2015	12	305853000	54
2016	1	699880000	57
2016	2	331513000	57
2016	3	309561608	52
2016	4	253940000	38
2016	5	339840000	51
2016	6	199669000	55

1a. Year over year funding growth

```sql
SELECT
    Year,
    SUM(Amount_USD) AS total_funding,
    LAG(SUM(Amount_USD)) OVER (ORDER BY Year) AS prev_year_funding,
    ROUND(
        (SUM(Amount_USD) - LAG(SUM(Amount_USD)) OVER (ORDER BY Year)) * 100.0
        / LAG(SUM(Amount_USD)) OVER (ORDER BY Year), 2
    ) AS yoy_growth_pct
FROM cleaned_startup_funding
GROUP BY Year
ORDER BY Year;
```

OUTPUT :-

Year	total_funding	prev_year_funding	yoy_growth_pct
2015	8673022368	NULL	NULL
2016	3828088608	8673022368	-55.86
2017	10409809730	3828088608	171.93
2018	5122368369	10409809730	-50.79
2019	9700918535.220001	5122368369	89.38
2020	390207254	9700918535.220001	-95.98

1b.Month over month funding growth

```sql
SELECT
    YearMonth,
    SUM(Amount_USD) AS total_funding,
    LAG(SUM(Amount_USD)) OVER (ORDER BY YearMonth) AS prev_month_funding,
    ROUND(
        (SUM(Amount_USD) - LAG(SUM(Amount_USD)) OVER (ORDER BY YearMonth)) * 100.0
        / LAG(SUM(Amount_USD)) OVER (ORDER BY YearMonth), 2
    ) AS mom_growth_pct
FROM cleaned_startup_funding
GROUP BY YearMonth
ORDER BY YearMonth;
```

OUTPUT :-

YearMonth	total_funding	prev_month_funding	mom_growth_pct
2015-01	370171768	NULL	NULL
2015-02	394262600	370171768	6.51
2015-03	463327000	394262600	17.52
2015-04	981000000	463327000	111.73
2015-05	326185000	981000000	-66.75
2015-06	536994000	326185000	64.63
2015-07	1606657000	536994000	199.19
2015-08	1063816000	1606657000	-33.79
2015-09	1470520000	1063816000	38.23
2015-10	254961000	1470520000	-82.66
2015-11	899275000	254961000	252.71
2015-12	305853000	899275000	-65.99
2016-01	699880000	305853000	128.83
2016-02	331513000	699880000	-52.63
2016-03	309561608	331513000	-6.62
2016-04	253940000	309561608	-17.97
2016-05	339840000	253940000	33.83
2016-06	199669000	339840000	-41.25

1c. Running cumulative funding over time

```
SELECT
    YearMonth,
    SUM(Amount_USD) AS monthly_funding,
    SUM(SUM(Amount_USD)) OVER (ORDER BY YearMonth) AS cumulative_funding
FROM cleaned_startup_funding
GROUP BY YearMonth
ORDER BY YearMonth;
```

OUTPUT :-

YearMonth	monthly_funding	cumulative_funding
2015-01	370171768	370171768
2015-02	394262600	764434368
2015-03	463327000	1227761368
2015-04	981000000	2208761368
2015-05	326185000	2534946368
2015-06	536994000	3071940368
2015-07	1606657000	4678597368
2015-08	1063816000	5742413368
2015-09	1470520000	7212933368
2015-10	254961000	7467894368
2015-11	899275000	8367169368
2015-12	305853000	8673022368
2016-01	699880000	9372902368
2016-02	331513000	9704415368
2016-03	309561608	10013976976
2016-04	253940000	10267916976
2016-05	339840000	10607756976
2016-06	199669000	10807425976

Industries by total funding

```
SELECT
    Industry,
    SUM(Amount_USD) AS total_funding,
    COUNT(*) AS total_deals,
    ROUND(AVG(Amount_USD), 2) AS avg_deal_size
FROM cleaned_startup_funding
GROUP BY Industry
ORDER BY total_funding DESC;
```

OUTPUT :-

Industry	total_funding	total_deals	avg_deal_size
Ecommerce	7163710608	170	42139474.16
Consumer Internet	6254084245	590	10600142.79
Transportation	3916632394	4	979158098.5
Technology	2229707930	310	7192606.23
Finance	1951938000	56	34856035.71
Unknown	1231811368	131	9403140.21
FinTech	1224211265	10	122421126.5
E-commerce	1097025084.54	35	31343573.84
Online Marketplace	700143000	2	350071500
E-Commerce & M-Commerce platform	680000000	1	680000000
B2B	587000000	2	293500000
Car Aggregator & Retail Mobile App	500000000	1	500000000
Ecommerce Marketplace	500000000	1	500000000
Health and wellness	490686000	6	81781000
Healthcare	418218000	45	9293733.33
Cab Aggregator	400000000	1	400000000
Education	301143000	19	15849631.58
Food and Beverages	300311200	3	100103733.33

2a. Industry funding by year, to see shift over time

```sql
SELECT
    Year,
    Industry,
    SUM(Amount_USD) AS total_funding
FROM cleaned_startup_funding
GROUP BY Year, Industry
ORDER BY Year, total_funding DESC;
```

OUTPUT :-

Year	Industry	total_funding
2015	Unknown	1231811368
2015	Online Marketplace	700000000
2015	E-Commerce & M-Commerce platform	680000000
2015	Car Aggregator & Retail Mobile App	500000000
2015	Ecommerce Marketplace	500000000
2015	Cab Aggregator	400000000
2015	Cab rental Mobile app	225000000
2015	Online Classifieds	150000000
2015	Logistics Solution Provider	137000000
2015	Online Food Delivery	121500000
2015	Hyper-local Grocery Delivery platform	120000000
2015	Mobile Advertising platform	100000000
2015	Budget Hotel accommodation brand	100000000
2015	Music Streaming App	100000000
2015	Budget Hotel Accommodation	100000000
2015	Online home d cor marketplace	100000000
2015	Doctor Appointment booking app	90000000
2015	Ecommerce Logistics	85000000

2b. Top 5 industries each year, ranked

```sql
SELECT *
FROM (
    SELECT
        Year,
        Industry,
        SUM(Amount_USD) AS total_funding,
        RANK() OVER (PARTITION BY Year ORDER BY SUM(Amount_USD) DESC) AS industry_rank
    FROM cleaned_startup_funding
    GROUP BY Year, Industry
) ranked
WHERE industry_rank <= 5
ORDER BY Year, industry rank;
```

OUTPUT :-

Year	Industry	total_funding	industry_rank
2015	Unknown	1231811368	1
2015	Online Marketplace	700000000	2
2015	E-Commerce & M-Commerce platform	680000000	3
2015	Car Aggregator & Retail Mobile App	500000000	4
2015	Ecommerce Marketplace	500000000	4
2016	Consumer Internet	1918775000	1
2016	eCommerce	975046608	2
2016	Technology	682513000	3
2016	Education	82565000	4
2016	healthcare	52400000	5
2017	Ecommerce	5902584000	1
2017	Consumer Internet	2595576000	2
2017	Technology	981921730	3
2017	Finance	529570000	4
2017	Logistics	184700000	5
2018	Consumer Internet	1739733245	1
2018	Finance	1086768000	2
2018	Technology	536273200	3

2c. Industry share of total funding per year (percentage)

```
SELECT
    Year,
    Industry,
    SUM(Amount_USD) AS industry_funding,
    ROUND(
        SUM(Amount_USD) * 100.0 / SUM(SUM(Amount_USD)) OVER (PARTITION BY Year), 2
    ) AS pct_share_of_year
FROM cleaned_startup_funding
GROUP BY Year, Industry
ORDER BY Year, pct_share_of_year DESC;
```

OUTPUT :-

Year	Industry	industry_funding	pct_share_of_year
2015	Unknown	1231811368	14.2
2015	Online Marketplace	700000000	8.07
2015	E-Commerce & M-Commerce platform	680000000	7.84
2015	Car Aggregator & Retail Mobile App	500000000	5.77
2015	Ecommerce Marketplace	500000000	5.77
2015	Cab Aggregator	400000000	4.61
2015	Cab rental Mobile app	225000000	2.59
2015	Online Classifieds	150000000	1.73
2015	Logistics Solution Provider	137000000	1.58
2015	Online Food Delivery	121500000	1.4
2015	Hyper-local Grocery Delivery platform	120000000	1.38
2015	Mobile Advertising platform	100000000	1.15
2015	Budget Hotel accommodation brand	100000000	1.15
2015	Music Streaming App	100000000	1.15
2015	Budget Hotel Accommodation	100000000	1.15
2015	Online home d cor marketplace	100000000	1.15
2015	Doctor Appointment booking app	90000000	1.04
2015	Ecommerce Logistics	85000000	0.98

- Biggest city hubs by funding

```
SELECT
    City,
    SUM(Amount_USD) AS total_funding,
    COUNT(*) AS total_deals,
    ROUND(AVG(Amount_USD), 2) AS avg_deal_size
FROM cleaned_startup_funding
GROUP BY City
ORDER BY total_funding DESC;
```

OUTPUT :-

City	total_funding	total_deals	avg_deal_size
Bengaluru	18532908863	592	31305589.3
Mumbai	4958575015	407	12183231
Gurugram	3878113657.54	244	15893908.43
New Delhi	3370871515	271	12438640.28
Noida	1283164000	57	22511649.12
Unknown	1271863868	135	9421213.84
Chennai	718767000	75	9583560
Pune	717582000	77	9319246.75
Menlo Park	450000000	1	450000000
Hyderabad	382176230	72	5308003.19
India	349500000	5	69900000
California	300000000	1	300000000
Kormangala	284000000	2	142000000
Faridabad	252000000	2	126000000
Tulangan	200000000	1	200000000
San Franci...	187000000	2	93500000
Jaipur	152735000	14	10909642.86
San Jose	135000000	1	135000000

3a. City funding trend by year

```sql
SELECT
    City,
    Year,
    SUM(Amount_USD) AS total_funding
FROM cleaned_startup_funding
GROUP BY City, Year
ORDER BY City, Year;
```

OUTPUT :-

City	Year	total_funding
Ahmedabad	2015	28365000
Ahmedabad	2016	37371000
Ahmedabad	2017	43200000
Ahmedabad	2018	11044811
Amritsar	2019	300000
Andheri	2019	5750000
Belgaum	2015	500000
Bengaluru	2015	2985088000
Bengaluru	2016	1044896108
Bengaluru	2017	7416604000
Bengaluru	2018	1550714595
Bengaluru	2019	5317247300
Bengaluru	2020	218358860
Bhopal	2015	1800000
Bhopal	2016	100000
Bhopal	2019	430665
Bhubaneswar	2018	23000000
Boston	2015	3000000

3b.Top city per industry

```sql
SELECT *
FROM (
    SELECT
        Industry,
        City,
        SUM(Amount_USD) AS total_funding,
        RANK() OVER (PARTITION BY Industry ORDER BY SUM(Amount_USD) DESC) AS city_rank
    FROM cleaned_startup_funding
    GROUP BY Industry, City
) ranked
WHERE city_rank = 1
ORDER BY total_funding DESC;
```

OUTPUT :-

Industry	City	total_funding	city_rank
Ecommerce	Bengaluru	6063764608	1
Transportation	Bengaluru	3900000000	1
Consumer Internet	Bengaluru	2414368500	1
Finance	Mumbai	1360000000	1
Unknown	Unknown	1231811368	1
FinTech	Noida	1000000000	1
Online Marketplace	Bengaluru	700000000	1
E-Commerce & M-Commerce platform	New Delhi	680000000	1
Technology	Mumbai	613141500	1
B2B	Bengaluru	585000000	1
Car Aggregator & Retail Mobile App	Bengaluru	500000000	1
Ecommerce Marketplace	New Delhi	500000000	1
Health and Wellness	Menlo Park	450000000	1
Cab Aggregator	Bengaluru	400000000	1
E-commerce	Bengaluru	350085860	1
Automation	California	300000000	1
Last Mile Transportation	Kormangala	283000000	1
Hospitality	Gurugram	250000000	1

Distribution across funding stages

```
SELECT
    Investment_Type_Clean,
    SUM(Amount_USD) AS total_funding,
    COUNT(*) AS total_deals,
    ROUND(SUM(Amount_USD) * 100.0 / SUM(SUM(Amount_USD)) OVER (), 2) AS pct_of_total_funding
FROM cleaned_startup_funding
GROUP BY Investment_Type_Clean
ORDER BY total_funding DESC;
```

OUTPUT :-

Investment_Type_Clean	total_funding	total_deals	pct_of_total_funding
Private Equity	27215368165	1076	71.39
Series B	4805195735	21	12.6
Series D	1481799000	12	3.89
Other	1205432095	11	3.16
Seed/Angel	1056281523	864	2.77
Series C	1044718394	14	2.74
Series A	247572000	30	0.65
Series G	231000000	1	0.61
Unknown	207629010.22	3	0.54
Debt Funding	185183174	28	0.49
Series H	150000000	1	0.39
Equity	132000000	3	0.35
Series F	90000000	2	0.24
Corporate Round	38080000	1	0.1
Series E	33000000	2	0.09
Mezzanine	1000000	1	0
Crowd Funding	155768	2	0

4a. Funding stage trend by year

```
SELECT
    Year,
    Investment_Type_Clean,
    SUM(Amount_USD) AS total_funding,
    COUNT(*) AS total_deals
FROM cleaned_startup_funding
GROUP BY Year, Investment_Type_Clean
ORDER BY Year, total_funding DESC;
```

OUTPUT :-

Year	Investment_Type_Clean	total_funding	total_deals
2015	Private Equity	8506315000	352
2015	Seed/Angel	166551600	300
2015	Crowd Funding	155768	2
2016	Private Equity	3711404000	293
2016	Seed/Angel	116684608	293
2017	Private Equity	10290166000	291
2017	Seed/Angel	74163730	154
2017	Debt Funding	45480000	10
2018	Private Equity	4184052500	134
2018	Series B	309000000	4
2018	Seed/Angel	222479780	97
2018	Equity	132000000	3
2018	Series D	77799000	2
2018	Series C	62000000	2
2018	Debt Funding	51145089	10
2018	Series A	43272000	8
2018	Other	36620000	3
2018	Series E	3000000	1

Top investors by number of deals

```sql
SELECT
    Investors_Name,
    COUNT(*) AS total_deals
FROM cleaned_startup_funding
GROUP BY Investors_Name
ORDER BY total_deals DESC
LIMIT 10;
```

OUTPUT :-

Investors_Name	total_deals
Undisclosed Investors	64
Unknown	20
Undisclosed Investor	20
Group of Angel Investors	16
Indian Angel Network	14
Sequoia Capital	13
Kalaari Capital	13
SAIF Partners	12
Accel Partners	10
Undisclosed	10

5a.Top investors by total amount invested

```sql
SELECT
    Investors_Name,
    SUM(Amount_USD) AS total_amount
FROM cleaned_startup_funding
GROUP BY Investors_Name
ORDER BY total_amount DESC
LIMIT 10;
```

OUTPUT :-

Investors_Name	total_amount
SoftBank Group	4312000000
WestBridge Capital	3952000000
Microsoft, eBay, Tencent Holdings	1400000000
Vijay Shekhar Sharma	1000150000
Steadview Capital, existing investors	700000000
Alibaba Group, Ant Financial	680000000
Unknown	624105951
SoftBank Vision Fund	601000000
Altimeter Capital, DST Global	585000000
Alibaba, Foxconn, SoftBank Group	500000000

Average and median deal size by industry

```
WITH ranked_industry AS (
    SELECT
        Industry,
        Amount_USD,
        ROW_NUMBER() OVER (PARTITION BY Industry ORDER BY Amount_USD) AS row_num,
        COUNT(*) OVER (PARTITION BY Industry) AS row_count
    FROM cleaned_startup_funding
    WHERE Amount_USD IS NOT NULL
)
SELECT
    Industry,
    ROUND(AVG(Amount_USD), 2) AS median_deal_size
FROM ranked_industry
WHERE row_num IN (FLOOR((row_count + 1) / 2), CEIL((row_count + 1) / 2))
GROUP BY Industry;
```

OUTPUT :-

Industry	median_deal_size
3D Printer Manufacturer	3000000
Accounting	5000000
Advertising & Marketing Community Networking ...	1000000
Advertising, Marketing	6000000
Aerospace	50000000
Affordable Education	650000
Affordable Hotel Booking Online	5000000
Agriculture	1619517.34
Agtech	140000000
Alternate Mobile Monetization platform	5000000
API Workflow platform	1000000
App Analytics platform	500000
App based Bus Pooling Services	20000000
App security testing platform	500000
Architectural Design & Consulting	500000
Artificial Intelligence	38080000
Artificial Intelligence platform	750000
Asset Financing platform	1100000

6a. Average deal size by industry, separate query for side by side comparison

```
SELECT
    Industry,
    ROUND(AVG(Amount_USD), 2) AS avg_deal_size
FROM cleaned_startup_funding
WHERE Amount_USD IS NOT NULL
GROUP BY Industry
ORDER BY avg_deal_size DESC;
```

OUTPUT :-

Industry	avg_deal_size
Transportation	979158098.5
E-Commerce & M-Commerce platform	680000000
Car Aggregator & Retail Mobile App	500000000
Ecommerce Marketplace	500000000
Cab Aggregator	400000000
Online Marketplace	350071500
Automation	300000000
B2B	293500000
B2B Platform	225000000
Cab rental Mobile app	225000000
E-Tech	200000000
Online Classifieds	150000000
Last Mile Transportation	141650000
Agtech	140000000
Logistics Solution Provider	137000000
Hospitality	125000000
FinTech	122421126.5
Hyper-local Grocery Delivery platform	120000000

6b.Average and median deal size by funding stage

```
WITH ranked_stage AS (
    SELECT
        Investment_Type_Clean,
        Amount_USD,
        ROW_NUMBER() OVER (PARTITION BY Investment_Type_Clean ORDER BY Amount_USD) AS row_num,
        COUNT(*) OVER (PARTITION BY Investment_Type_Clean) AS row_count
    FROM cleaned_startup_funding
    WHERE Amount_USD IS NOT NULL
)
```

Investment_Type_Clean	median_deal_size
Corporate Round	38080000
Crowd Funding	77884
Debt Funding	4900000
Equity	31000000
Mezzanine	1000000
Other	6240000
Private Equity	6000000
Seed/Angel	368000
Series A	3450000
Series B	15800000
Series C	50000000
Series D	65000000
Series E	16500000
Series F	45000000
Series G	231000000
Series H	150000000
Unknown	4889975.54

```
SELECT
    Investment_Type_Clean,
    ROUND(AVG(Amount_USD), 2) AS median_deal_size
FROM ranked_stage
WHERE row_num IN (FLOOR((row_count + 1) / 2), CEIL((row_count + 1) / 2))
GROUP BY Investment_Type_Clean;
```

Investment_Type_Clean	median_deal_size
Corporate Round	38080000
Crowd Funding	77884
Debt Funding	4900000
Equity	31000000
Mezzanine	1000000
Other	6240000
Private Equity	6000000
Seed/Angel	368000
Series A	3450000
Series B	15800000
Series C	50000000
Series D	65000000
Series E	16500000
Series F	45000000
Series G	231000000
Series H	150000000
Unknown	4889975.54

```sql
SELECT
    Investment_Type_Clean,
    ROUND(AVG(Amount_USD), 2) AS avg_deal_size
FROM cleaned_startup_funding
WHERE Amount_USD IS NOT NULL
GROUP BY Investment_Type_Clean
ORDER BY avg_deal_size DESC;
```

Investment_Type_Clean	avg_deal_size
Series G	231000000
Series B	228818844.52
Series H	150000000
Series D	123483250
Other	109584735.91
Series C	74622742.43
Unknown	69209670.07
Series F	45000000
Equity	44000000
Corporate Round	38080000
Private Equity	25293093.09
Series E	16500000
Series A	8252400
Debt Funding	6613684.79
Seed/Angel	1222548.06
Mezzanine	1000000
Crowd Funding	77884

EXTRA ANALYTICAL QUESTIONS (BEYOUND THE 6 CORE QUESTIONS)

Overalldisclosedvsundiscloseddealpercentagees LECT

SUM(CASE WHEN Amount_USD IS NOT NULL THEN 1 ELSE 0 END) AS disclosed_deals,

---

SUM(CASE WHEN Amount_USD IS NULL THEN 1 ELSE 0 END) AS undisclosed_deals, ROUND(

SUM(CASE WHEN Amount_USD IS NOT NULL THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
) AS disclosed_pct
FROM cleaned_startup_funding;

Top 10 largest single funding rounds

SE LECT
Startup_Name,

Industry,
City,
Investors_Name,
Investment_Type_Clean,
Amount_USD,
Date

FROM cleaned_startup_funding WHERE
Amount_USD IS NOT NULL ORDER BY
Amount_USD DESC LIMIT 10;

□ Startups that raised more than one round, with total raised

```
SEL ECT
 Startup_Name,

   COUNT(*) AS total_rounds,
   SUM(Amount_USD) AS total_raised

FROM cleaned_startup_funding
GROUP BY Startup_Name HAVING
COUNT(*) > 1
ORDER BY total_raised DESC;
```

□ Top subverticals by funding within top industries

```
SEL ECT

   Industry,
   SubVertical,
   SUM(Amount_USD) AS total_funding,
   COUNT(*) AS total_deals

FROM cleaned_startup_funding
WHERE Industry IN (
   SELECT Industry FROM (
      SELECT Industry, SUM(Amount_USD) AS total
      FROM cleaned_startup_funding
      GROUP BY Industry
      ORDER BY total DESC
      LIMIT 5

   ) top_industries
)
GROUP BY Industry, SubVertical
ORDER BY Industry, total_funding DESC;
```

□ City and industry combined view, top 15 pairs by funding

```
SEL ECT
 City,

   Industry,
   SUM(Amount_USD) AS total_funding,
   COUNT(*) AS total_deals

FROM cleaned_startup_funding
GROUP BY City, Industry ORDER BY
total_funding DESC LIMIT 15;
```

□Number of unique investors and unique startups per year

```
SELECT
   Year,
   COUNT(DISTINCT Startup_Name) AS unique_startups,
   COUNT(DISTINCT Investors_Name) AS unique_investors,
   COUNT(*) AS total_deals

FROM cleaned_startup_funding
GROUP BY Year
ORDER BY Year;
```